Quantitative Methods for Management

Tutorial Exercise Sheet: Change of Basis and Eigenvalues

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Recap: Linear maps, kernel, and image. A map $\phi : \mathbb{R}^n \to \mathbb{R}^m$ assigns an output vector in $\mathbb{R}^m$ to each input vector in $\mathbb{R}^n$. It is *linear* if it preserves addition and scalar multiplication:

$$\begin{aligned} \phi(\mathbf{u}+\mathbf{v}) &= \phi(\mathbf{u}) + \phi(\mathbf{v}) \quad \text{(additivity)}, \\ \phi(\lambda\mathbf{u}) &= \lambda\phi(\mathbf{u}) \qquad\quad \text{(homogeneity)}. \end{aligned}$$

Both identities must hold for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and all $\lambda \in \mathbb{R}$. Thus, adding or scaling inputs has the same effect as adding or scaling their outputs. We denote the canonical basis by $\mathbb{E}$ and other ordered bases by $\mathbb{F}$ or $\mathbb{G}$. The notation $\mathbf{x}_\mathbb{F}$ denotes the column of coordinates of $\mathbf{x}$ in the basis $\mathbb{F}$. For fixed bases $\mathbb{F}$ of $\mathbb{R}^n$ and $\mathbb{G}$ of $\mathbb{R}^m$, a linear map has the representation $\phi(\mathbf{x})_\mathbb{G} = \mathbf{A}_\phi \mathbf{x}_\mathbb{F}$, where $\mathbf{A}_\phi$ is the unique $m \times n$ matrix representing ϕ in these bases. In the following, 0 denotes the zero vector in the relevant space.

For a linear map, the *kernel* contains all inputs sent to zero, and the *image* contains all outputs that can be reached:

$$\begin{aligned} \ker \phi &= \{\mathbf{x} \in \mathbb{R}^n : \phi(\mathbf{x}) = 0\}, \\ \operatorname{im} \phi &= \{\phi(\mathbf{x}) : \mathbf{x} \in \mathbb{R}^n\}. \end{aligned}$$

The kernel is a subspace of the domain $\mathbb{R}^n$. The image is a subspace of the codomain $\mathbb{R}^m$. For a linear map ϕ, we always have:

$$\dim \ker \phi + \dim \operatorname{im} \phi = n.$$

The number of independent directions "lost" plus the number of independent output directions equals the dimension of the domain. In particular, $\dim \operatorname{im} \phi \leq \min\{n, m\}$.

To analyze a map in the exercises below, we proceed as follows:

- **Linearity check.** First, compute $\phi(0)$. If $\phi(0) \neq 0$, the map is not linear. If $\phi(0) = 0$, continue: This condition alone does not prove linearity! Proceed to verify additivity and homogeneity by checking both identities above using arbitrary vectors and scalars. If both hold in general, the map is linear.
- **Finding a basis of the kernel.** Solve $\phi(\mathbf{x}) = 0$, or equivalently $\mathbf{A}_\phi \mathbf{x}_\mathbb{F} = 0$. Express the solution using free parameters, as the coefficient vectors form a basis. The number of basis vectors is $\dim \ker \phi$. If the only solution is $\mathbf{x} = 0$, the kernel has the empty basis and dimension zero.
- **Finding a basis of the image.** Write $\phi(\mathbf{x})$ as a linear combination of fixed vectors with the input coordinates as coefficients. These vectors span the image; keep a linearly independent subset that still spans it. Equivalently, row-reduce $\mathbf{A}_\phi$ and take the pivot columns of the *original* matrix as a basis. The number of basis vectors is $\dim \operatorname{im} \phi$. If the image is $\{0\}$, its basis is empty and its dimension is zero.

- **Sanity check with the Theorem of Dimensions.** As a last sanity check, use the Theorem of Dimensions to verify that the dimensions of the kernel and image are correct.

Exercise 1 (Linearity, kernel, and image)**.** For each map, determine whether it is linear and justify your answer. If it is linear, find a basis for its kernel and its image, and state their dimensions. If it is not linear, give a counterexample to one of the linearity conditions.

(a) $\phi_1 : \mathbb{R}^2 \to \mathbb{R}^2$, defined by $\phi_1(x, y) = (x + y,\ 2x + 2y)$.

(b) $\phi_2 : \mathbb{R}^3 \to \mathbb{R}^2$, defined by $\phi_2(x, y, z) = (x + z,\ y + z)$.

(c) $\phi_3 : \mathbb{R}^2 \to \mathbb{R}^3$, defined by $\phi_3(x, y) = (x,\ y,\ xy)$.

(d) $\phi_4 : \mathbb{R}^2 \to \mathbb{R}^2$, defined by $\phi_4(x, y) = (x + y + 1,\ x - y)$.

Recap: The matrix of a linear map and changes of basis. In the following, we work in $\mathbb{R}^n$ and want to understand how to change between the representations in the canonical basis $\mathbb{E}$ and one other ordered basis $\mathbb{F} = (\mathbf{f}_1, \ldots, \mathbf{f}_n)$. We denote as $\mathbf{x}_\mathbb{F}$ the coefficients needed to express the vector $\mathbf{x}$ as a linear combination of the vectors in $\mathbb{F}$. We obtain the change-of-basis matrix by placing the canonical coordinates of these basis vectors in columns:

$$\mathbf{P} = \begin{pmatrix} (\mathbf{f}_1)_\mathbb{E} & \cdots & (\mathbf{f}_n)_\mathbb{E} \end{pmatrix}, \qquad \mathbf{x}_\mathbb{E} = \mathbf{P}\mathbf{x}_\mathbb{F}, \qquad \mathbf{x}_\mathbb{F} = \mathbf{P}^{-1}\mathbf{x}_\mathbb{E}.$$

The matrix $\mathbf{P}$ transforms a vector from its representation in the basis $\mathbb{F}$ into canonical coordinates. For example, in two dimensions, the coordinate column $(a, b)^T$ means that we take a times the first basis vector and b times the second. Multiplication by $\mathbf{P}$ forms exactly this combination:

$$\mathbf{x} = a\mathbf{f}_1 + b\mathbf{f}_2, \qquad \mathbf{x}_\mathbb{E} = \mathbf{P}\begin{pmatrix} a \\ b \end{pmatrix} = a(\mathbf{f}_1)_\mathbb{E} + b(\mathbf{f}_2)_\mathbb{E}.$$

Conversely, $\mathbf{P}^{-1}$ recovers the coefficients a and b from canonical coordinates. The vector stays fixed. $\mathbf{P}$ and $\mathbf{P}^{-1}$ only its representation. Since the vectors of $\mathbb{F}$ are linearly independent, $\mathbf{P}$ is invertible. Dependent columns cannot define a basis or give unique coordinates for every vector.

To convert coordinates, we proceed as follows:

- **Form the matrix.** We place the vectors of $\mathbb{F}$ in columns, in their given order, using canonical coordinates.
- **Choose the direction.** From $\mathbb{F}$ to $\mathbb{E}$, we multiply by $\mathbf{P}$. From $\mathbb{E}$ to $\mathbb{F}$, we multiply by $\mathbf{P}^{-1}$, or solve $\mathbf{P}\mathbf{x}_\mathbb{F} = \mathbf{x}_\mathbb{E}$.
- **Check the result.** We reconstruct the vector from its basis coefficients and verify that its canonical coordinates agree with the original ones.

The columns also give a geometric interpretation of $\mathbf{P}$: as a linear transformation of canonical coordinates, it sends each canonical basis vector to the corresponding vector of $\mathbb{F}$. For example, in $\mathbb{R}^2$, consider

$$\mathbb{F} = ((2, 0), (1, 1)), \qquad \mathbf{P} = \mathbf{A}_\phi = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \phi(x, y) = (2x + y, y).$$

The map $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ stretches and shears the canonical grid, with $\phi(\mathbf{e}_1) = \mathbf{f}_1$ and $\phi(\mathbf{e}_2) = \mathbf{f}_2$. The picture below shows this transformation. When we instead use $\mathbf{P}$ to convert coordinates, it describes the same vector using a different basis; it does not move that vector.

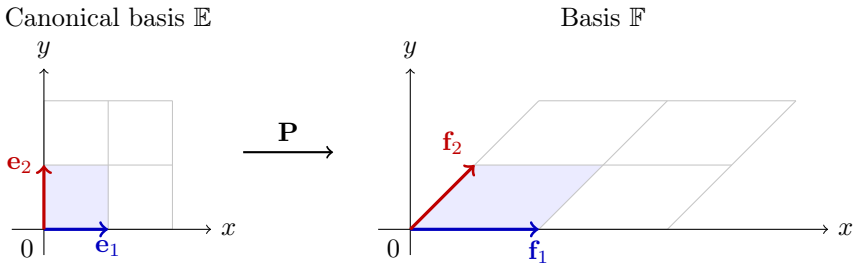


Figure 1: The combined map $\phi(x, y) = (2x + y, y)$ sends the canonical grid directly to the stretched and sheared grid. The colored arrows show the basis vectors and their images; the shaded unit square becomes a parallelogram.

For this basis, conversion from canonical coordinates to $\mathbb{F}$ gives

$$\mathbf{x}_{\mathbb{F}} = \mathbf{P}^{-1}\mathbf{x}_{\mathbb{E}} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (x-y)/2 \\ y \end{pmatrix}.$$

Indeed, $((x-y)/2)(2,0) + y(1,1) = (x, y)$, so these coefficients reconstruct the original vector.

Exercise 2 (Change of basis)**.** Let $\mathbb{E}$ be the canonical basis of $\mathbb{R}^2$, and let $\mathbb{F} = ((1,0),(1,1))$.

(a) Which matrix converts coordinates from $\mathbb{F}$ to $\mathbb{E}$, so that $\mathbf{x}_{\mathbb{E}} = \mathbf{P}\mathbf{x}_{\mathbb{F}}$? Justify your choice using the columns of the matrix.

$$\mathbf{P}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{P}_2 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{P}_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{P}_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(b) Find $\mathbf{P}^{-1}$ and state which coordinate conversion it performs.

(c) For $\mathbf{x}_{\mathbb{E}} = (2,1)^T$, compute $\mathbf{x}_{\mathbb{F}}$. Reconstruct the vector using the vectors of $\mathbb{F}$ and verify your answer.

(d) Could the ordered pair $((1,1),(2,2))$ replace $\mathbb{F}$ as a basis of $\mathbb{R}^2$? Explain your answer.

Recap: Evaluating a linear map given in another basis. Let $\phi : \mathbb{R}^n \to \mathbb{R}^n$ be linear. Suppose $\mathbf{M}$ represents the map ϕ in the basis $\mathbb{F}$. This means that

$$\phi(\mathbf{x})_{\mathbb{F}} = \mathbf{M}\mathbf{x}_{\mathbb{F}}.$$

As before, $\mathbf{P}$ has the vectors of $\mathbb{F}$ as columns in canonical coordinates. If the input is given in the canonical basis $\mathbb{E}$, we first express it in $\mathbb{F}$, apply $\mathbf{M}$, and then convert the output back to $\mathbb{E}$:

$$\mathbf{x}_{\mathbb{E}} \xrightarrow{\mathbf{P}^{-1}} \mathbf{x}_{\mathbb{F}} \xrightarrow{\mathbf{M}} \phi(\mathbf{x})_{\mathbb{F}} \xrightarrow{\mathbf{P}} \phi(\mathbf{x})_{\mathbb{E}}.$$

- **Convert the input.** We compute $\mathbf{x}_{\mathbb{F}} = \mathbf{P}^{-1}\mathbf{x}_{\mathbb{E}}$.

- **Apply the map.** We compute $\phi(\mathbf{x})_{\mathbb{F}} = \mathbf{M}\mathbf{x}_{\mathbb{F}}$. The result is still expressed in $\mathbb{F}$.
- **Convert the output.** We compute $\phi(\mathbf{x})_{\mathbb{E}} = \mathbf{P}\phi(\mathbf{x})_{\mathbb{F}}$.

The coordinate conversions leave the input and output vectors unchanged, as the just change the representations of the vector. The matrix then $\mathbf{M}$ performs the calculation of the map in the basis $\mathbb{F}$. Combining the three steps gives

$$\phi(\mathbf{x})_{\mathbb{E}} = \mathbf{P}\mathbf{M}\mathbf{P}^{-1}\mathbf{x}_{\mathbb{E}}, \qquad \mathbf{A}_{\phi} = \mathbf{P}\mathbf{M}\mathbf{P}^{-1}.$$

The product is read from right to left. We can evaluate a vector one step at a time without first multiplying the three matrices. The matrix $\mathbf{P}$ must be invertible because it changes basis. The matrix $\mathbf{M}$ need not be invertible.

Exercise 3 (Evaluating a linear map in another basis)**.** Let $\mathbb{E}$ be the canonical basis of $\mathbb{R}^2$, and let $\mathbb{F} = ((1,1),(0,1))$. A linear map $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ is represented in the basis $\mathbb{F}$ by

$$\mathbf{M} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \phi(\mathbf{x})_{\mathbb{F}} = \mathbf{M}\mathbf{x}_{\mathbb{F}}.$$

(a) Construct the matrix $\mathbf{P}$ that converts coordinates from $\mathbb{F}$ to $\mathbb{E}$, and find its inverse.

(b) For $\mathbf{x}_{\mathbb{E}} = (1,2)^T$, compute $\mathbf{x}_{\mathbb{F}}$.

(c) Compute $\phi(\mathbf{x})_{\mathbb{F}}$ using $\mathbf{M}$, then convert the result to $\phi(\mathbf{x})_{\mathbb{E}}$.

(d) Compute $\mathbf{A}_{\phi} = \mathbf{P}\mathbf{M}\mathbf{P}^{-1}$ and use it to verify your result from (c).

Recap: Eigenvalues, multiplicities, and eigenspaces. Let $\mathbf{A}$ be a real $n \times n$ matrix. A real number λ is an *eigenvalue* if there is a nonzero vector $\mathbf{v}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}, \qquad \text{or equivalently} \qquad (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0,$$

where $\mathbf{I}$ is the $n \times n$ identity matrix. The vector $\mathbf{v}$ is an *eigenvector*. The map scales it by λ: positive eigenvalues preserve its direction, negative eigenvalues reverse it, and $\lambda = 0$ sends it to zero. The zero vector is never an eigenvector.

We use the convention

$$p_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I})$$

for the *characteristic polynomial*. Its roots are the eigenvalues. The *algebraic multiplicity* $m_a(\lambda_0)$ counts how many times an eigenvalue λ_0 occurs as a root: it is the exponent of the factor $(\lambda_0 - \lambda)$ in the fully factored polynomial. A real matrix need not have only real eigenvalues but here we restrict our attention to matrices whose eigenvalues are all real.

For an eigenvalue λ, the *eigenspace* is

$$\mathcal{V}_{\lambda} = \ker(\mathbf{A} - \lambda\mathbf{I}) = \{\mathbf{v} \in \mathbb{R}^n : (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0\}.$$

It contains all eigenvectors for λ, together with the zero vector. Its dimension is the *geometric multiplicity*, $m_g(\lambda) = \dim \mathcal{V}_{\lambda}$. Thus, algebraic multiplicity is read from a polynomial, whereas geometric multiplicity is found by solving a linear system.

The *trace* of a matrix is the sum of its diagonal entries. It also equals the sum of its eigenvalues, counted with their algebraic multiplicities:

$$\mathrm{Tr}(\mathbf{A}) = \sum_{j=1}^{n} a_{jj} = \lambda_1 + \cdots + \lambda_n.$$

If all but one of these eigenvalues are known, we can find the remaining one by subtracting the known eigenvalues from the trace. Repeated eigenvalues must be included as many times as their algebraic multiplicity indicates.

We proceed as follows:

- **Look for useful structure.** If $\mathbf{A}$ is upper or lower triangular, its eigenvalues are its diagonal entries, since the characteristic polynomial is of form $p_{\mathbf{A}}(\lambda) = \prod_{j=1}^{n}(a_{jj} - \lambda)$. Repeated diagonal entries give repeated roots. If $\mathbf{A}$ is singular, that is, not invertible, then 0 is an eigenvalue because $\mathbf{A}\mathbf{v} = 0$ has a nonzero solution. Conversely, an eigenvalue of 0 implies that $\mathbf{A}$ is singular. Its eigenspace is $\mathcal{V}_0 = \ker \mathbf{A}$; singularity alone does not determine its multiplicities.

- **Find the roots and their algebraic multiplicities.** We first form $\mathbf{A} - \lambda\mathbf{I}$ by subtracting λ from each diagonal entry and leaving the other entries unchanged. For a 2×2 matrix, we calculate its determinant by multiplying the diagonal entries and subtracting the product of the off-diagonal entries:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc.$$

  We expand the products, collect powers of λ, and set the determinant equal to zero:

$$(a - \lambda)(d - \lambda) - bc = 0.$$

  We then solve this equation by factoring or using the quadratic formula. Its roots are the eigenvalues, and the number of times each root occurs gives its algebraic multiplicity. For a 3×3 matrix, we can calculate the determinant by expanding along a row or column, preferably one containing zeros. A known root λ_0 of the resulting cubic allows us to factor out $(\lambda_0 - \lambda)$ and solve the remaining quadratic. If the matrix is singular, we already know the root 0 and can first factor out λ.

- **Find each eigenspace.** For every distinct eigenvalue, we solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = 0$ and write the solution using free parameters. Their coefficient vectors form a basis of $\mathcal{V}_\lambda$. The number of basis vectors is $m_g(\lambda)$. For a triangular matrix, we still need to solve this system as repeated diagonal entries alone do not determine the geometric multiplicity.

- **Check the results.** Each proposed basis vector must satisfy $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. For every eigenvalue,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

  Since all eigenvalues considered here are real, their algebraic multiplicities must add up to n. A simple root has both multiplicities equal to one. A repeated root may have a smaller geometric multiplicity than its algebraic multiplicity.

Exercise 4 (Eigenvalues, multiplicities, and eigenspaces)**.** For each matrix, find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace. Identify any useful structure before computing the characteristic polynomial. All eigenvalues are real.

(a) $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

(b) $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

(c) $\mathbf{C} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

(d) $\mathbf{D} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

(e) $\mathbf{E} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

First find $\ker \mathbf{E}$. Then use its dimension and the trace of $\mathbf{E}$ to determine all eigenvalues and their multiplicities without expanding the characteristic polynomial. Find a basis for each eigenspace.